

15.2 Determining Sample Space

Lesson Introduction

 Benchmark	MA.7.DP.2.1 Determine the sample space for a simple experiment.		 Learning Target	I can determine the sample space for a simple experiment.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.8.DP.2.1 Determine the sample space for a repeated experiment.	
 Essential Questions	- How do the number of possible outcomes for an experiment relate to the likelihood of a particular event? - How can we apply probability to inform a prediction?			
 Lesson Narrative	In this lesson, students begin to assign probabilities to chance events and identify the sample spaces for simple experiments. They build the understanding that the greater the probability, the more likely an event is to occur. Furthermore, students determine whether outcomes in a simple experiment are “equally likely,” building on the concept of fairness explored in the previous lesson.			
 Component Summary	15.2.1 Warm-Up (~5 min)	Students choose between two games to play (SE pg. 406)	15.2.3 Bring It Together (5-10 min)	Students reflect on exploration and apply new vocabulary (SE pg. 409)
	15.2.2 Exploration (15-20 min)	Deeper exploration into games from the Warm-Up and discovery of sample space (SE pg. 407)	15.2.4 Guided Practice (10 min)	Practice finding the sample space in a simple experiment (SE pg. 409)
	15.2.5 Wrap-Up (~5 min)	Formative assessment on finding the sample space in a simple experiment (SE pg. 410)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Simple experiment - Probability <u>Additional Terms:</u> - Sample space		- Letter cards for 15.2.2 Exploration (Optional) Probability manipulatives- coins, dice, spinners	N/A

 Teacher Tips	Common Misconceptions - Students may think that the phrase “equally likely” means that there is a 50% chance of an outcome happening. - Students may think that if a specific outcome has not occurred when they attempted an experiment, it means that outcome is impossible.	Things to Consider In 15.2.2 Exploration, model understanding of “equally likely” outcomes by changing the questions to ask about different outcomes in the sample space. For example, ask about the probability of rolling a 1 on a die, then a 4, then a 5, etc.
	Literacy and Language Routines Think-Pair-Share 15.2.3 Bring It Together is a summary activity of the partner Exploration. In the activity, students are asked to consider the likelihood of different outcomes from the paper activity. Allow each student to make a conjecture about the answer to the questions and share with a partner to explain their reasoning. Listen as students are sharing to support in summarizing the key takeaways of the exploration.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.4.1: Discussion and Reflection 15.2.2 Exploration is a collaboration opportunity for students. Students get to work together to compare the likelihood of winning between two games. They will analyze the mathematical thinking of others and justify their reasoning, gaining important practice in this Mathematical Thinking and Reasoning Standard.
	 Differentiate Instruction The simple experiments explored in 15.2.1 Warm-Up and 15.2.2 Exploration contain examples of common probability games. Consider preparing sets of manipulatives (dice, coins, and spinners). The use of manipulatives offers kinesthetic and visual entry points for learning about outcomes and sample spaces.	
Lesson Notes:		

15.2.1 Warm-Up: Which Game Would You Choose?

Component Overview: Students analyze two games to determine which game they would rather play. In this Warm-Up, students apply their innate understanding of probability.



15.2.1 Warm-Up: Which Game Would You Choose?

1. Two games are described.

Game 1: A player flips a coin and wins if it lands on heads.

Game 2: A player rolls a standard number cube and wins if it lands on a number that is divisible by 3.

- a. Which game would you rather play?

Answers will vary. I would rather play Game 1. I would rather play Game 2.

- b. Explain your reasoning.

I would rather play Game 1 because my chances of winning are equal to my chances of losing. I would rather play Game 2 because there is more than one chance of winning.



Use this Warm-Up as a way to gauge students' understanding about probability from their experience playing games. Students do not need to use formalized language of probability to answer this question. Allow students to share their responses, but do not provide a "correct" answer.

15.2.2 Exploration: Simple Experiments

Component Overview: Working with partners, students make connections between a simple game's sample space, possible outcomes, and the probability of winning. Students review the example games from the Warm-Up and discover how to prove mathematically which game has a greater likelihood of a player winning. In the second half of the Exploration, students are introduced to the idea that not all sample spaces are obvious before actually performing the experiment.



15.2.2 Exploration: Simple Experiments

Work with your partner to complete the following.

1. Discuss the sample space for both games from the Warm-Up with your partner.
 - a. Write the sample space for each in the table.

Random Process	Sample Space
Flipping a Coin 	heads, tails
Rolling a Standard Number Cube 	1, 2, 3, 4, 5, 6

- b. What are the possible outcomes in the event of landing on heads in Game 1 from the Warm-Up?
Heads
 - c. What fraction of the possible outcomes from Game 1 results in a win?
 $\frac{1}{2}$
 - d. What are the possible outcomes in the event of rolling a number divisible by 3 in Game 2 from the Warm-Up?
three, six
 - e. What fraction of the possible outcomes from Game 2 results in a win?
 $\frac{1}{3}$



For a kinesthetic and visual approach to this activity, provide partner pairs with a coin and a number cube. This allows students to visualize the sample space by completing trials of the experiment.



How did you figure out the sample space for each random process?

15.2.2 Exploration: Simple Experiments (continued)

- f. Based on your answers to Parts B-E, complete the statement.

A player has a greater likelihood of winning

☐ Game 1
☐ Game 2

because $\frac{1}{2} > \frac{1}{3}$.

- g. How could Game 2 be modified so the likelihood of winning either game is the same?

Answers will vary. Game 2 can be modified so that a player can win if the standard number cube lands on an even number.

2. With your partner, determine who is partner A and who is partner B. For this activity, you and your partner will pull slips of paper from a bag.

- a. Make a guess about what is on the slips of paper.
Answers will vary. I guess numbers are on the slips of paper.
- b. Without looking in the bag, partner A should take out one of the papers and show it to partner B. Both partners record what was printed on the paper in the first row of the table.
Answers will vary.

Who?	What is printed on the paper?
Partner A, Choice 1	
Partner B, Choice 1	
Partner A, Choice 2	
Partner B, Choice 2	
Partner A, Choice 3	
Partner B, Choice 3	



MA.K12.MTR.4.1: Discussion and Reflection

Consider having a whole class discussion as to why the larger fraction indicates a greater likelihood of winning. This could illuminate potential misconceptions on how students understand probability.

Ideally, students will understand that the closer the fraction is to 1, which means that winning is certain, the more likely that game is to produce a win.



For an additional challenge, ask students if Game 1 could be modified to match the likelihood of Game 2.



Provide each partner pair with a paper bag containing the cut-up letter strips from the Blackline Master.



Demonstrate the steps for the activity by acting out an example round. This can help students process how the activity works before attempting it.

15.2.2 Exploration: Simple Experiments (continued)

- c. Partner A should put the paper back into the bag. Then, partner B should take out one of the papers and show it to partner A for both to record in the table.
- d. After both partners have made their first selection and put it back in the bag, make a revised guess of the sample space.
Answers will vary. I guess letters are on the slips of paper.
- e. Continue taking turns and recording the results in the table until each partner has taken three turns.
- f. Make a revised guess of the sample space.
Answers will vary. I guess letters are on the slips of paper.
- g. Rate your confidence in your last guess using the emoji scale.
Answers will vary.



- h. Discuss with your partner what you could do to make a better guess of the sample space. Write at least two strategies you discussed.
Answers will vary. I can take all of the papers out of the bag. I can also ask another classmate.
- i. Look at all the papers in the bag. Write the sample space.
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



- What would be helpful to make a more accurate guess of the sample space?
- Can you tell if all outcomes are equally likely?
- How did you refine your sample space guesses for each round?
- Were any of your guesses correct for the sample space?
- What information would have helped you make a better guess?



If time allows, consider facilitating a longer discussion about outcomes and sample spaces. Ask students to think about having a new bag of papers and taking out a strip 100 times. If they never pulled out the letter "Q," would this necessarily mean that "Q" was not a part of the sample space? What could they conclude?

15.2.3 Bring It Together: Defining Simple Experiments

Component Overview: Students are introduced to formal vocabulary “simple experiment” and “probability.” Students confirm the games they explored previously are simple experiments and compare the likelihood of different outcomes in one of the games.



15.2.3 Bring It Together: Defining Simple Experiments

The games from the exploration are all considered **simple experiments**.

A **simple experiment** is an action that has a definite outcome that cannot be predicted with certainty.

In a simple experiment, each outcome in the sample space is equally likely to happen; however, events, which are subsets of the sample space, may have different likelihoods of happening. This is called probability.

The **probability** of an event is a measure of the likelihood the event will occur.

1. Consider the simple experiment involving pulling papers from the bag in the Exploration.
 - a. Explain whether all of the possible outcomes in the sample space were equally likely or not.
Answers will vary. All of the possible outcomes were equally likely because there is only one of each letter.
 - b. Explain which of the following events would have been more likely to happen in that experiment – pulling out a paper with a vowel on it or pulling out a paper with a consonant on it.
Answers will vary. Pulling out a paper with a consonant on it would have been more likely because there are more consonants than there are vowels.



- What does it mean to say that “each outcome in the sample space is equally likely to happen?”
- Can you give an example of an experiment where each outcome is not equally likely to happen?



Think-Pair-Share

For question 1b, consider using a think-pair-share activity. Allow students independent thinking time to consider the two different outcomes and write a response. Then, they can share their thinking with a partner and revise their written responses if needed. Students will then share out as a whole group.

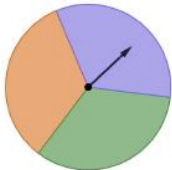
15.2.4 Guided Practice: Determining Sample Space of Simple Experiments

Component Overview: Students practice finding the sample space of simple experiments and applying their knowledge of probability to create a fair game.



15.2.4 Guided Practice: Determining Sample Space of Simple Experiments

- For each simple experiment in the table, determine the sample space.

Simple Experiment	Sample Space
Spinning the spinner 	<i>orange, purple, green</i>
Rolling a fair 12-sided die 	<i>1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12</i>



Students can complete the first task independently. Consider starting the second task independently and allowing students to check their work with a partner before reviewing as a whole group.

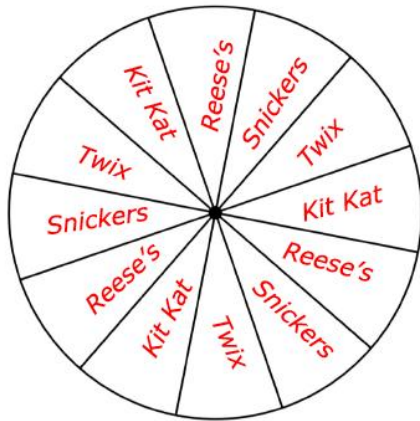


- Are the outcomes in these experiments “equally likely?”

15.2.4 Guided Practice: Determining Sample Space of Simple Experiments (continued)

2. Complete the following.

- a. Given the sample space of {Reese's, Snickers, Twix, Kit Kat}, label the game wheel so that there is a fair chance of winning each candy bar as a prize.



- b. Explain how you determined the labels for the game board.

Answers will vary. I divided the number of sections on the game wheel, 12, by the number of outcomes in the sample spaces, 4. I labeled the wheel with each outcome 3 times to be sure that all outcomes have equal chances of winning.



Consider scaffolding 2a to assist students who may have struggled in 15.2.2 Exploration and 15.2.3 Bring It Together.

Suggested questions:

- How many outcomes are on the spinner?
- How many outcomes are in the sample space?
- What does it mean for there to be a fair chance of winning each candy bar?



Provide an opportunity for meaningful reflection with question 2b. Allowing students to hear one another's thought processes can solidify their thinking on what it means for outcomes to be "equally likely." This response could be coupled with the Wrap-Up for you to evaluate student progress from the lesson.

15.2.5 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a formative assessment on identifying the sample space of a simple experiment.



15.2.5 Wrap-Up: Ticket Out the Door

1. For each simple experiment in the table, determine the sample space.

Simple Experiment	Sample Space
Selecting a day of the week from a deck of cards with each day written on one card	<i>Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday</i>
Flipping an integer chip 	<i>yellow, red</i>



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.



Consider asking students to compare the likelihood of the outcomes in the given simple experiments. Could students modify the experiments so that the outcomes are not “equally likely?”